

实验四

每个线程对应矩阵 C 中的一个元素，首先计算该线程负责的元素在矩阵中的位置：

```
// Calculate the row index of the P element and M
// *** TO DO: Compute the row index for the current thread ***
// int row = ...;
int row = blockIdx.y * blockDim.y + threadIdx.y;

// Calculate the column index of the P element and N
// *** TO DO: Compute the column index for the current thread ***
// int col = ...;
int col = blockIdx.x * blockDim.x + threadIdx.x;
```

接着，遍历矩阵 A 和 B 的相关行列，按照矩阵乘法的规则进行计算，并将结果累加到 pValue 中：

```
// Each thread computes one element of the matrix
// *** TO DO: Implement the matrix multiplication for a single element ***
for (int k = 0; k < width; k++) {
    pValue += d_M[row * width + k] * d_N[k * width + col];
}
```

最后，每个线程需要将计算得到的结果 pValue 写回到矩阵 C 的相应位置。在 CUDA 中，矩阵 C 是存储在全局内存中的，每个线程必须使用其计算得到的 row 和 col 索引来存储值：

```
// Store the computed value into the output matrix
// *** TO DO: Write the computed value to the correct position in d_P ***
// d_P[row * width + col] = ...;
d_P[row * width + col] = pValue;
```

测试正确性：

```
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 1 1000
Kernel Elapsed Time: 123.051 ms
Performance= 16.25 GFlop/s, Time= 123.051 msec, Size= 2000000000 Ops
Computing result using host CPU...done.
Listing first 100 Differences > 0.000010...

Total Errors = 0
```

块 block 为 16，测试结果：

```
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 1000
Kernel Elapsed Time: 6.640 ms
Performance= 301.19 GFlop/s, Time= 6.640 msec, Size= 2000000000 Ops
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 3000
Kernel Elapsed Time: 155.943 ms
Performance= 346.28 GFlop/s, Time= 155.943 msec, Size= 5400000000 Ops
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 5000
Kernel Elapsed Time: 732.482 ms
Performance= 341.31 GFlop/s, Time= 732.482 msec, Size= 25000000000 Ops
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 10000
Kernel Elapsed Time: 6312.411 ms
Performance= 316.84 GFlop/s, Time= 6312.411 msec, Size= 200000000000 Ops
(base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 20000
+Kernel Elapsed Time: 126314.547 ms
Performance= 126.67 GFlop/s, Time= 126314.547 msec, Size= 1600000000000 Ops
```

块 block 为 32, 测试结果:

```
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 1000
Kernel Elapsed Time: 6.941 ms
Performance= 288.14 GFlop/s, Time= 6.941 msec, Size= 2000000000 Ops
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 3000
Kernel Elapsed Time: 156.360 ms
Performance= 345.36 GFlop/s, Time= 156.360 msec, Size= 5400000000 Ops
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 5000
Kernel Elapsed Time: 730.747 ms
Performance= 342.12 GFlop/s, Time= 730.747 msec, Size= 25000000000 Ops
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 10000
Kernel Elapsed Time: 6339.227 ms
Performance= 315.50 GFlop/s, Time= 6339.227 msec, Size= 200000000000 Ops
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 20000
Kernel Elapsed Time: 75635.398 ms
Performance= 211.54 GFlop/s, Time= 75635.398 msec, Size= 1600000000000 Ops
```

实验五

首先, 每个线程负责将矩阵A和矩阵B中的一个元素加载到共享内存中的对应位置, 这里在加载时进行边界检查:

```
// Load the matrices from device memory
// to shared memory; each **thread** loads
// one element of each matrix
// --- TO DO :Load the elements of the sub-matrix of A into As ---
// ---          Load the elements of the sub-matrix of B into Bs ---
// NOTE: Ensure that the thread indices do not exceed the matrix dims
//       Use boundary checks to load valid elements into shared memory,
As[ty][tx] = (a / wA + ty < wA && a % wA + tx < wA) ?
    A[a + ty * wA + tx] : 0.0f;

Bs[ty][tx] = (b / wB + ty < wB && b % wB + tx < wB) ?
    B[b + ty * wB + tx] : 0.0f;
```

接着, 每个线程通过遍历共享内存中的矩阵元素, 计算相应的矩阵C的元素:

```
// --- TO DO :Implement the matrix multiplication using the sub-mat
for (int k = 0; k < BLOCK_SIZE; ++k) {
    Csub += As[ty][k] * Bs[k][tx];
}
```

最后, 每个线程将计算结果写回全局内存中的矩阵C:

```
// --- TO DO :Store the computed Csub result into matrix C ---
// NOTE: Ensure that the thread indices "c" do not exceed the m
//       Use boundary checks to write valid elements to the out
int row = by * BLOCK_SIZE + ty;
int col = bx * BLOCK_SIZE + tx;

if (row < wA && col < wB) {
    C[row * wB + col] = Csub;
}
```

正确性测试:

```
• (base) jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 1 1000
Kernel Elapsed Time: 141.921 ms
Performance= 14.09 GFlop/s, Time= 141.921 msec, Size= 2000000000 Ops
Computing result using host CPU...done.
Listing first 100 Differences > 0.000010...

Total Errors = 0
```

块 block 大小为 16, 测试结果:

- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 1000`
Kernel Elapsed Time: 6.094 ms
Performance= 328.19 GFlop/s, Time= 6.094 msec, Size= 2000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 2000`
Kernel Elapsed Time: 43.407 ms
Performance= 368.60 GFlop/s, Time= 43.407 msec, Size= 16000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 1000`
Kernel Elapsed Time: 5.976 ms
Performance= 334.66 GFlop/s, Time= 5.976 msec, Size= 2000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 3000`
Kernel Elapsed Time: 136.615 ms
Performance= 395.27 GFlop/s, Time= 136.615 msec, Size= 54000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 5000`
Kernel Elapsed Time: 630.421 ms
Performance= 396.56 GFlop/s, Time= 630.421 msec, Size= 25000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 10000`
Kernel Elapsed Time: 5081.161 ms
Performance= 393.61 GFlop/s, Time= 5081.161 msec, Size= 200000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 20000`
Kernel Elapsed Time: 84991.820 ms
Performance= 188.25 GFlop/s, Time= 84991.820 msec, Size= 1600000000000 Ops

块 block 大小为 32，测试结果：

- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 1000`
Kernel Elapsed Time: 157.612 ms
Performance= 12.69 GFlop/s, Time= 157.612 msec, Size= 2000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 3000`
Kernel Elapsed Time: 159.515 ms
Performance= 338.53 GFlop/s, Time= 159.515 msec, Size= 54000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 5000`
Kernel Elapsed Time: 711.947 ms
Performance= 351.15 GFlop/s, Time= 711.947 msec, Size= 250000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 10000`
Kernel Elapsed Time: 5667.335 ms
Performance= 352.90 GFlop/s, Time= 5667.335 msec, Size= 2000000000000 Ops
- (base) `jzd@DESKTOP-75609C8:~/lab4-5$ ./a.out 0 20000`
Kernel Elapsed Time: 62372.000 ms
Performance= 256.53 GFlop/s, Time= 62372.000 msec, Size= 16000000000000 Ops